



Digital Signal Processing

Lecture 4 Ideal Digital Filters

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DTFT– Sinusoids

- Recall that

$$x[n] = 1 \Leftrightarrow X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi r)$$

- Also note that

$$x[n]e^{j\omega_0 n} \Leftrightarrow X(e^{j(\omega - \omega_0)})$$

- Therefore

$$e^{j\omega_0 n} \Leftrightarrow \sum_{r=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi r)$$

$$\cos\omega_0 n \Leftrightarrow \sum_{r=-\infty}^{\infty} \pi\delta(\omega + \omega_0 + 2\pi r) + \pi\delta(\omega - \omega_0 + 2\pi r)$$

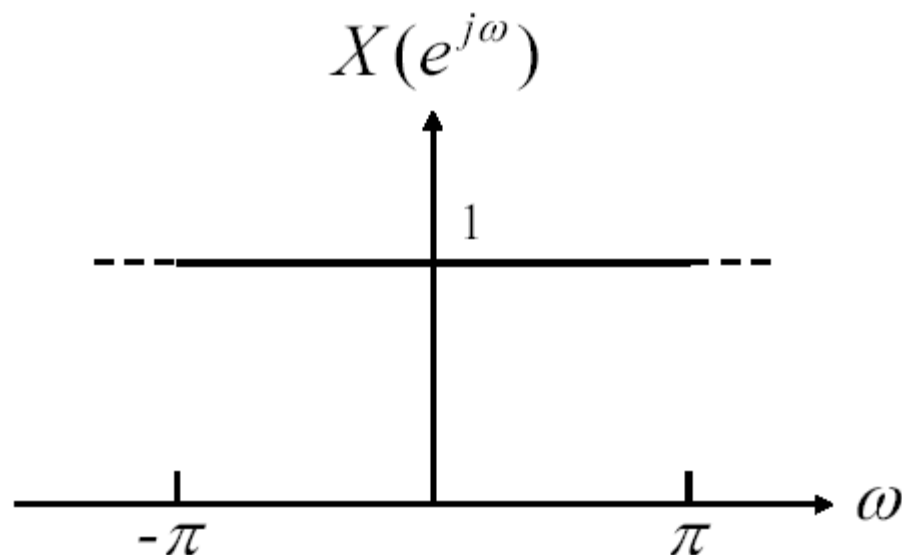
DTFT of Unit Impulse

Direct Transform

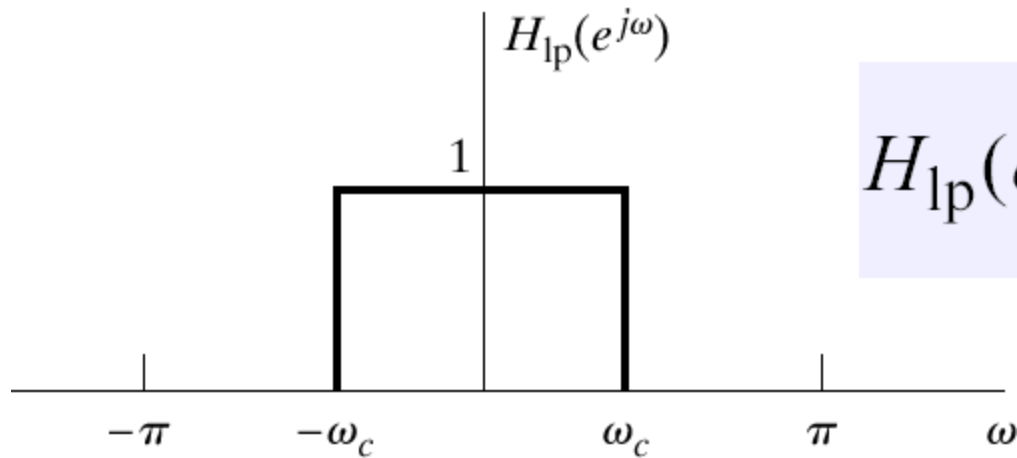
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n} = 1 \quad \text{for all } \omega$$

Inverse Transform

$$\delta[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 \times e^{j\omega n} d\omega$$



Ideal Lowpass Filter



$$H_{lp}(e^{j\omega}) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

$$\begin{aligned} h_{lp}[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_{lp}(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega \\ &= \frac{1}{j2\pi n} \left[e^{j\omega n} \right]_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{j2\pi n} \end{aligned}$$

$$h_{lp}[n] = \frac{\sin \omega_c n}{\pi n} \quad -\infty < n < \infty$$

Example

- Consider the real exponential signal

$$x[n] = a^n u[n]$$

- Its DTFT is obtained as follows:

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}$$

$$X(e^{j\omega}) = \frac{1 - ae^{j\omega}}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

$$= \underbrace{\frac{1 - a \cos \omega}{1 + a^2 - 2a \cos \omega}}_{X_R(e^{j\omega})} + j \underbrace{\frac{-a \sin \omega}{1 + a^2 - 2a \cos \omega}}_{X_I(e^{j\omega})}$$

Magnitude and Angle Form

$$x[n] = a^n u[n] \quad \Leftrightarrow \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})}$$

$$|X(e^{j\omega})|^2 = X(e^{j\omega}) X^*(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

$$= \frac{1}{1 + a^2 - 2a \cos \omega}$$

$$\angle X(e^{j\omega}) = \arctan\left(\frac{-a \sin \omega}{1 - a \cos \omega}\right)$$

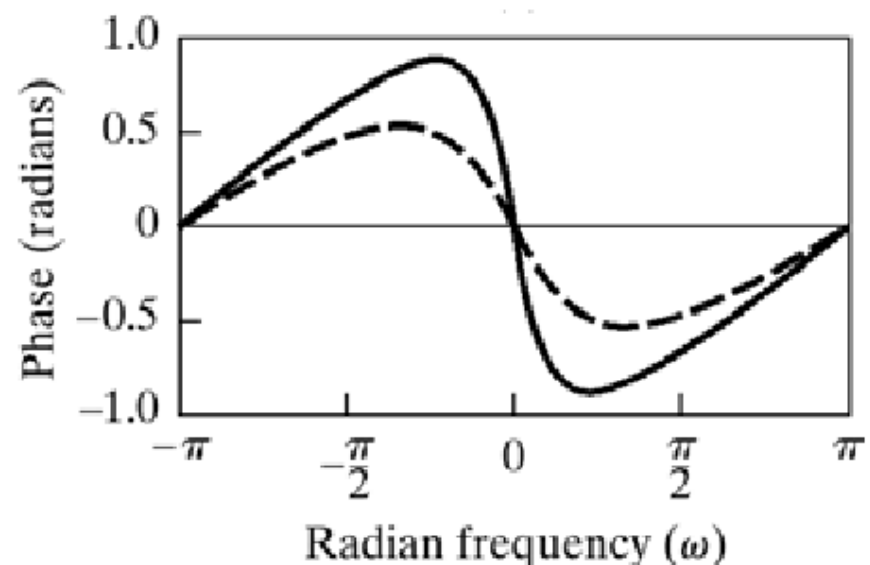
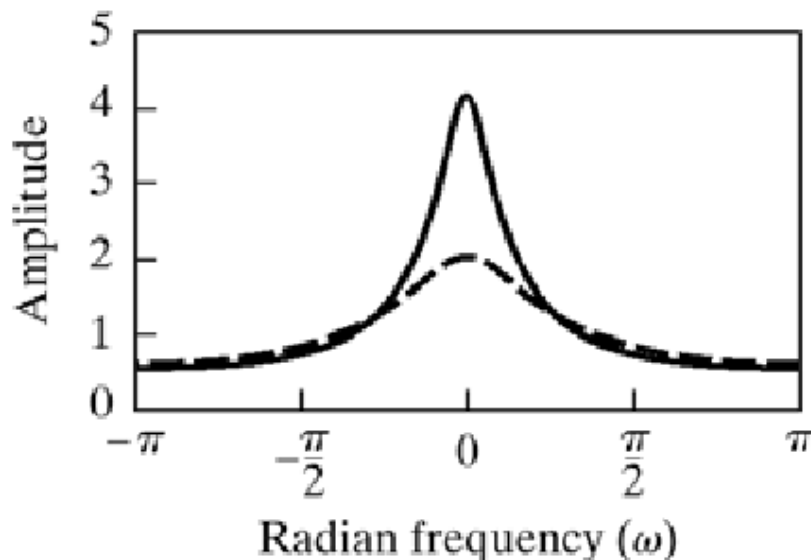
Magnitude and Angle Plot

$$|X(e^{j\omega})| = \frac{1}{(1 + a^2 - 2a \cos \omega)^{1/2}}$$

$$\angle X(e^{j\omega}) = \arctan\left(\frac{-a \sin \omega}{1 - a \cos \omega}\right)$$

$$|X(e^{j\omega})| = |X(e^{-j\omega})|$$

$$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$$



Example

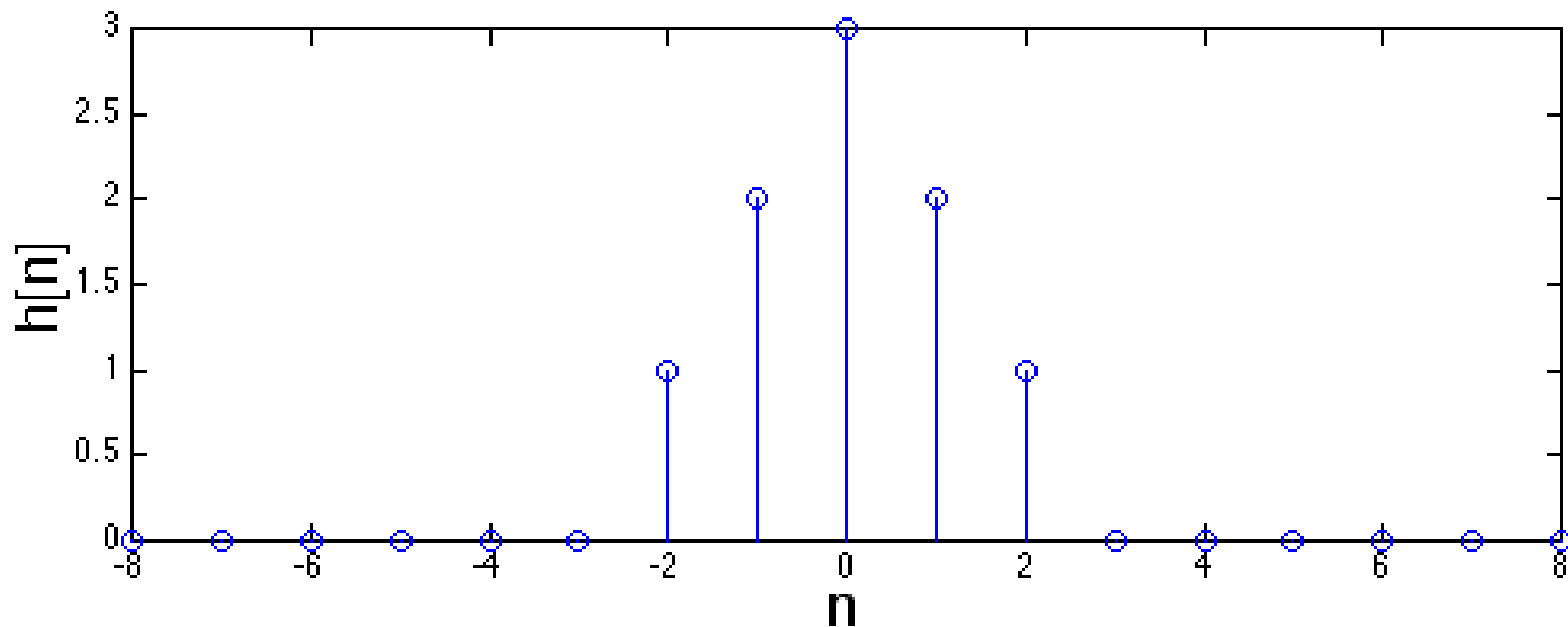
Determine $X(e^{j\omega})$ if $x[n] = a^{|n|}$ and $|a| < 1$.

Easy to do using the time reversal theorem

$$\begin{aligned}X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} a^{|n|} e^{-j\omega n} \\&= \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n} + \sum_{n=-\infty}^{\infty} a^{-n} u[-n-1] e^{-j\omega n} \\&= \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n} + a \sum_{n=-\infty}^{\infty} a^{-n-1} u[-n-1] e^{-j\omega n} \\&= \frac{1}{1 - ae^{-j\omega}} + \frac{ae^{j\omega}}{1 - ae^{j\omega}} \\&= \frac{1 - a^2}{1 - 2a \cos \omega + a^2}\end{aligned}$$

Real and Even (Zero Phase)

Consider an LTI system with an even unit sample response



$$= e^{2j\omega} + 2e^{j\omega} + 3 + 2e^{-j\omega} + e^{-2j\omega}$$

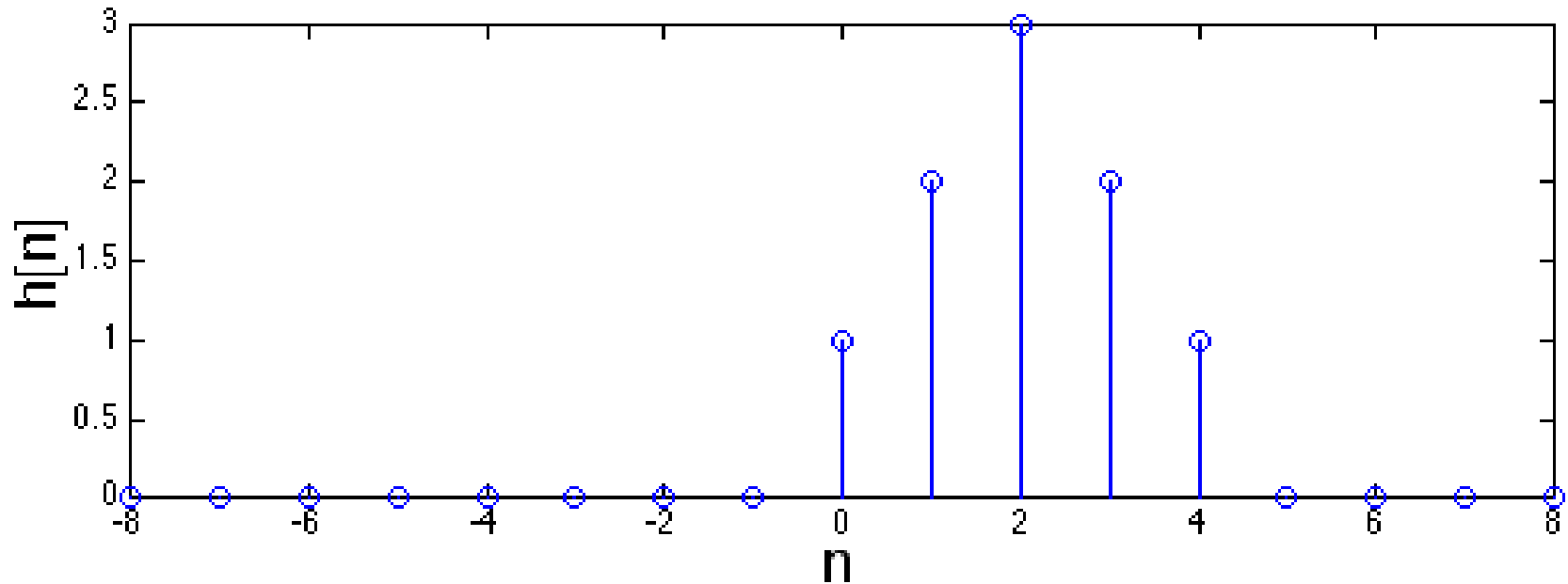
DTFT is

$$= 2\cos(2\omega) + 4\cos(\omega) + 3$$

Real & Even (Zero Phase)

- Frequency response is real, so system has “zero” phase shift
- This is to be expected since unit sample response is real and even.

Linear Phase



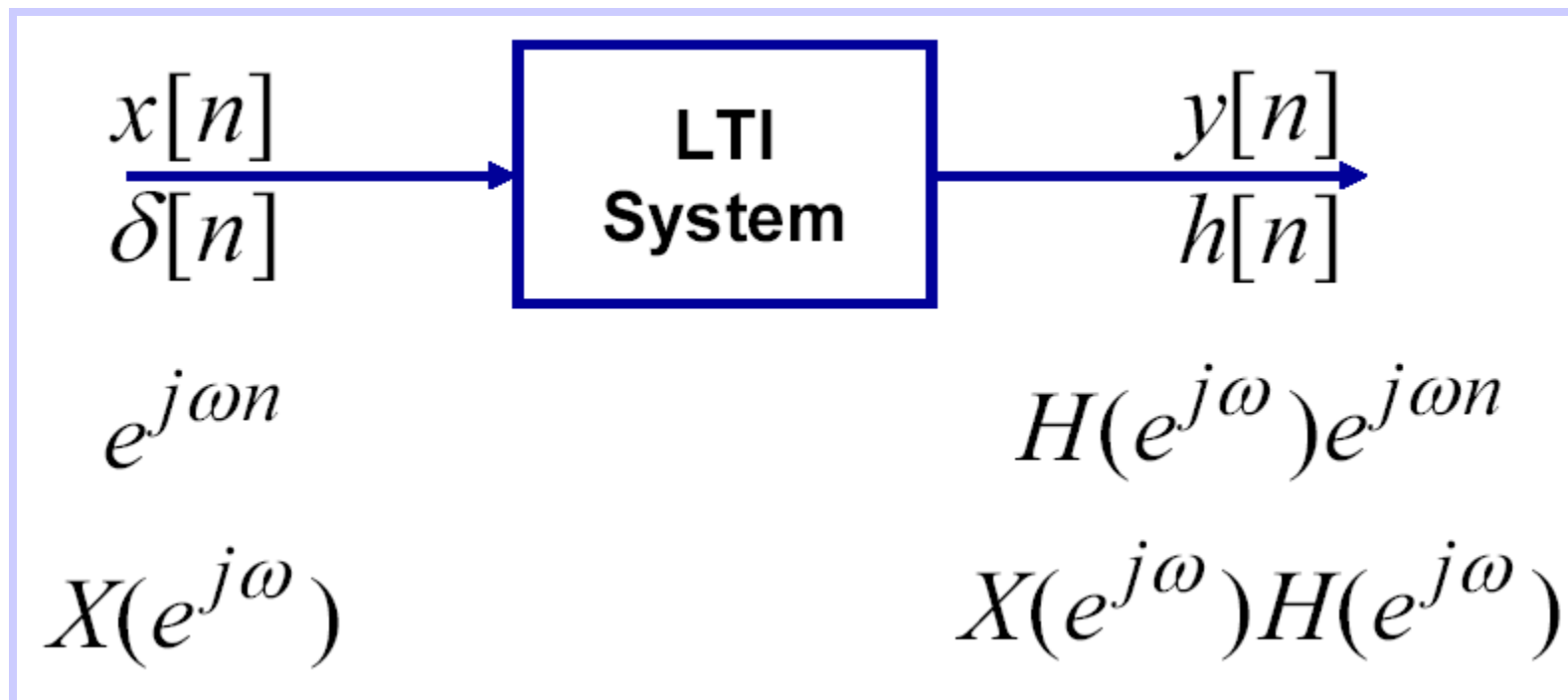
$$\begin{aligned} H(e^{j\omega}) &= e^{2j\omega} + 2e^{j\omega} + 3 + 3e^{-j\omega} + e^{-2j\omega} \\ &= e^{-2j\omega} (e^{2j\omega} + 2e^{j\omega} + 3 + 2e^{-j\omega} + e^{-2j\omega}) \\ &= e^{-2j\omega} (2\cos(2\omega) + 4\cos(\omega) + 3) \end{aligned}$$

Useful DTFT Pairs

- unit sample: $\delta[n] \xleftrightarrow{\mathcal{F}} 1$
- constant sequence: $1 \xleftrightarrow{\mathcal{F}} 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega + 2\pi k)$
- complex exponential: $e^{j\omega_0 n} \xleftrightarrow{\mathcal{F}} 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \omega_0 + 2\pi k)$
- sinusoid: $\cos \omega_0 n \xleftrightarrow{\mathcal{F}} \pi \left[\sum_{k=-\infty}^{+\infty} \delta(\omega - \omega_0 + 2\pi k) + \delta(\omega + \omega_0 + 2\pi k) \right]$
- unit step: $u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{+\infty} \delta(\omega + 2\pi k)$
- exponential sequence: $a^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1 - ae^{-j\omega}}, \text{ when } |a| < 1$

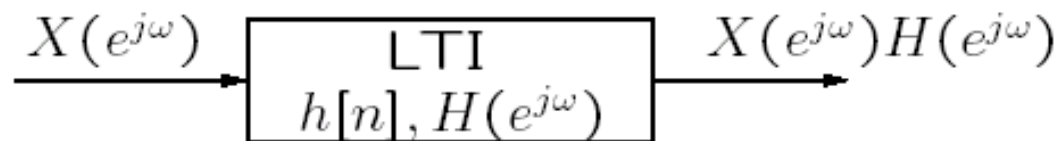
Convolution Theorem

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \Leftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

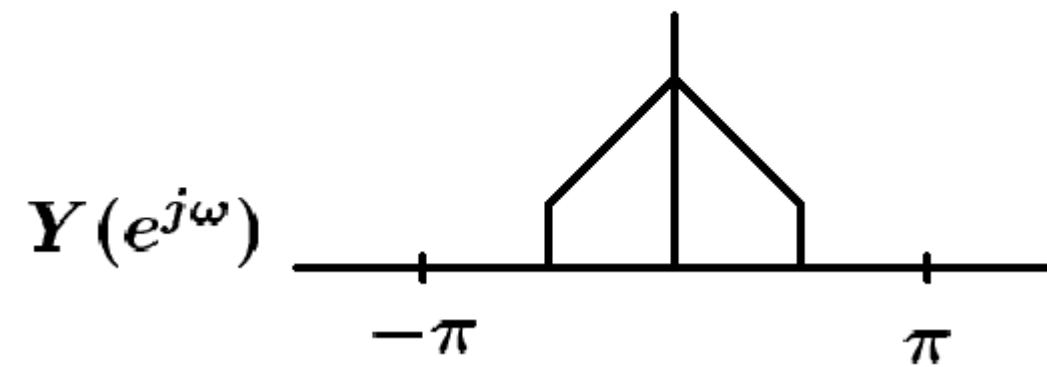
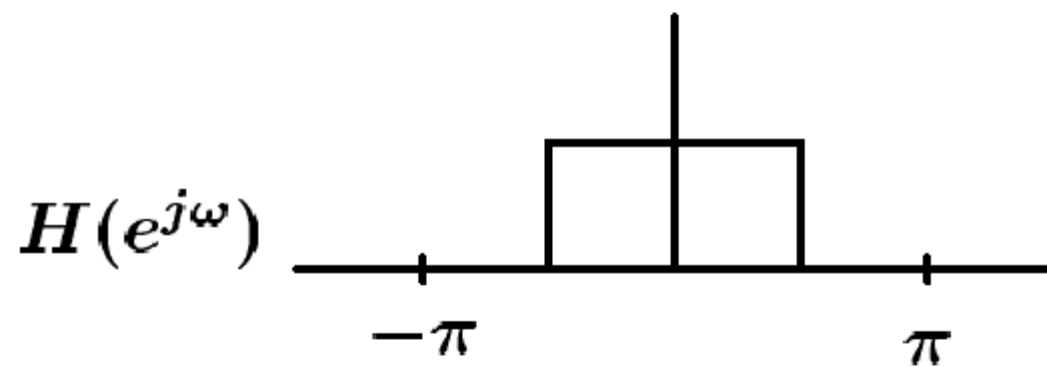
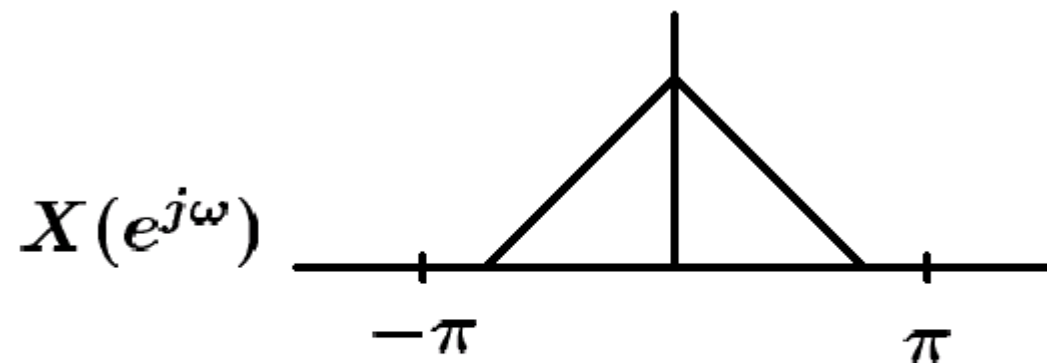


Linear Phase... (cont.)

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$



- Each frequency ω is affected by $H(e^{j\omega})$
 - gain or attenuation of $|H(e^{j\omega})|$
 - phase rotation of $\angle H(e^{j\omega})$
- the output is then $\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) H(e^{j\omega}) e^{j\omega n} d\omega$



Frequency Response of DE

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$\sum_{k=0}^N a_k Y(e^{j\omega}) e^{-j\omega k} = \sum_{k=0}^M b_k X(e^{j\omega}) e^{-j\omega k}$$

$$\left(\sum_{k=0}^N a_k e^{-j\omega k} \right) Y(e^{j\omega}) = \left(\sum_{k=0}^M b_k e^{-j\omega k} \right) X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\left(\sum_{k=0}^M b_k e^{-j\omega k} \right)}{\left(\sum_{k=0}^N a_k e^{-j\omega k} \right)}$$

Matlab

»help freqz

FREQZ Z-transform digital filter frequency response.

When N is an integer, $[H,W] = \text{FREQZ}(B,A,N)$ returns the N-point frequency vector W in radians and the N-point complex frequency response vector H given numerator and denominator coefficients in vectors B and A. The frequency response is evaluated at N points equally spaced around the upper half of the unit circle. If N isn't specified, it defaults to 512.

```
>> omega=(0:400)*pi/400; b=[1,1,1,1,1]/5;
```

```
>> H=freqz(b,1,omega);
```

```
>> subplot(211); plot(omega/pi,abs(H))
```

```
>> subplot(212); plot(omega/pi,angle(H))
```

Matlab

$[H,W] = \text{FREQZ}(B,A,N)$ gives N-point complex frequency response vector H and the N-point frequency vector W in radians/sample

$H = \text{FREQZ}(B,A,W)$ gives frequency response at vector W

$[H,F] = \text{FREQZ}(B,A,N,F_s)$ return frequency vector F (in Hz)

$H = \text{FREQZ}(B,A,F,F_s)$ gives complex frequency response at vector F in Hz

$\text{FREQZ}(B,A,...)$ with no output arguments plots the magnitude and unwrapped phase of the filter in the current figure window.

Example

- Suppose that the difference equation is
$$y[n] = y[n-1] - .9y[n-2] + x[n] + x[n-1]$$
- The frequency response is

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega}}{1 - e^{-j\omega} + .9e^{-j\omega 2}}$$

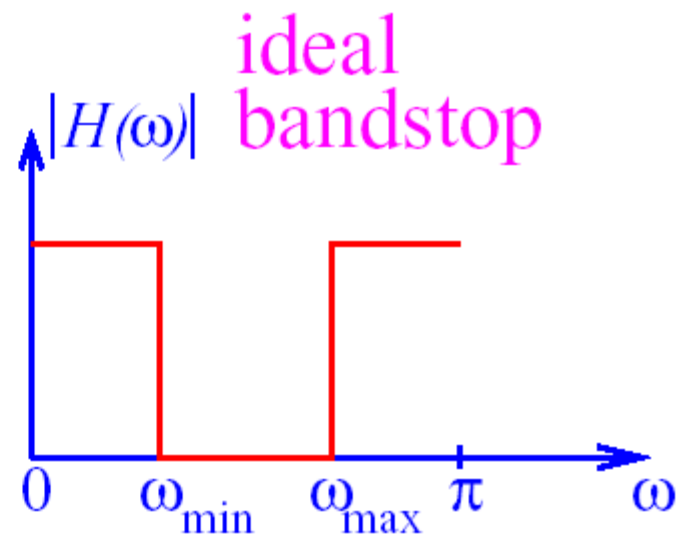
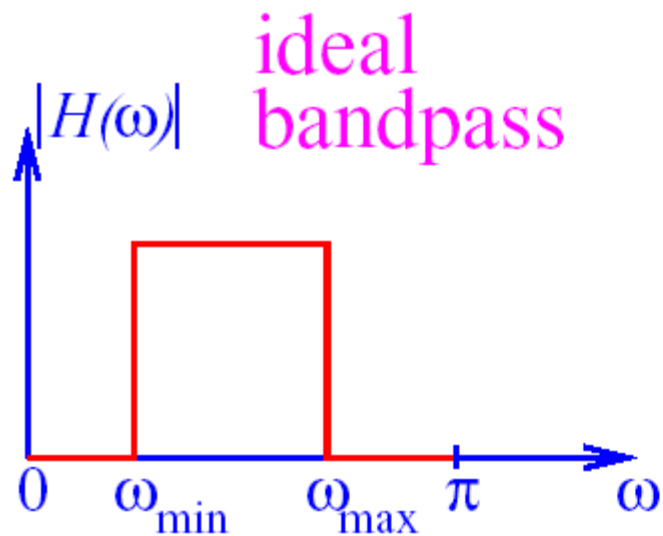
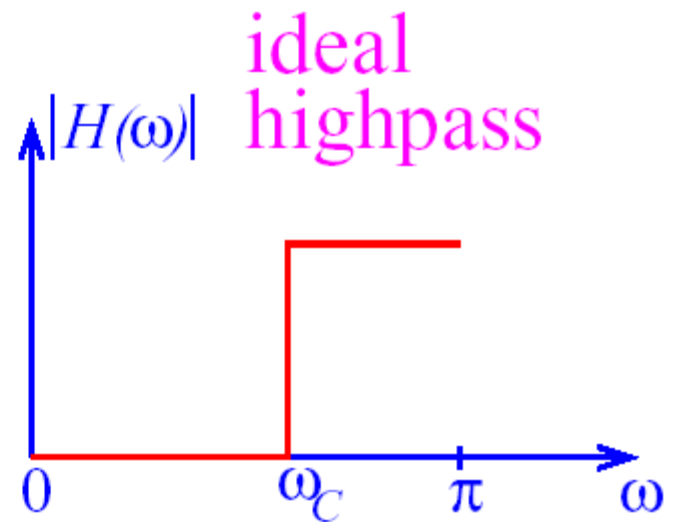
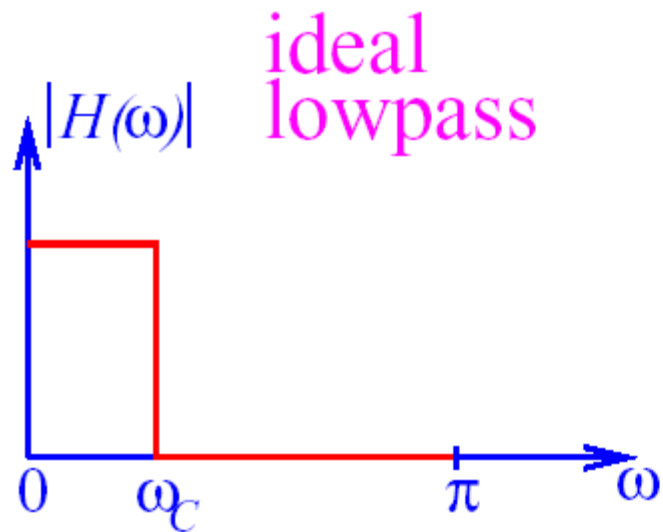
- This system is implemented in MATLAB by

```
>> y=filter([1,1],[1,-1,.9],x)
```

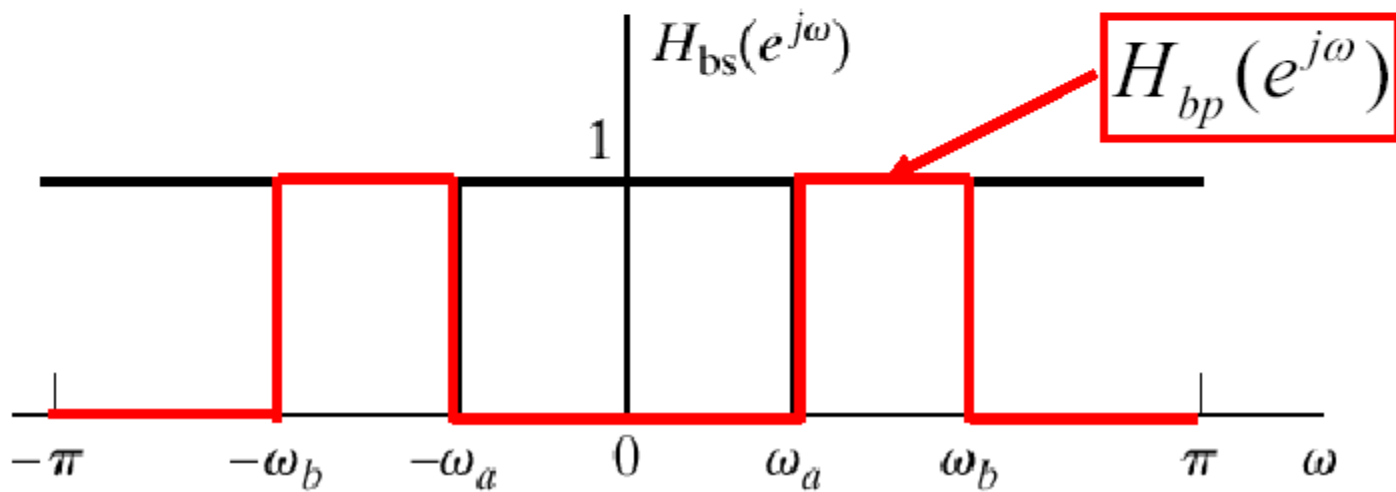
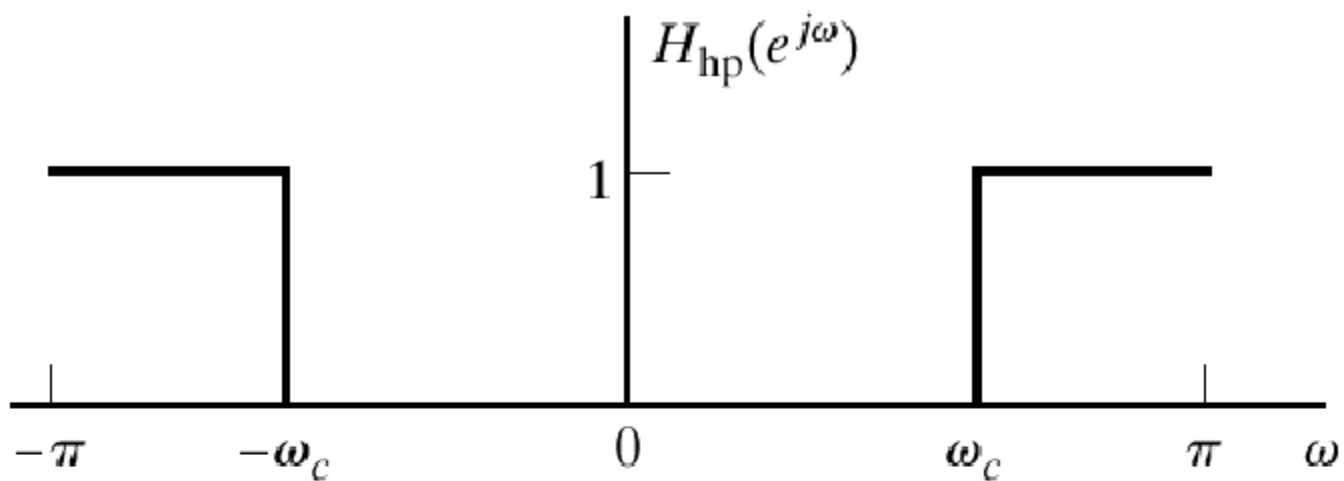
We can compute its frequency response by

```
>> omega=(0:500)*pi/500;  
>> H=freqz ([1,1],[1,-1,.9],omega);
```

Ideal Filters

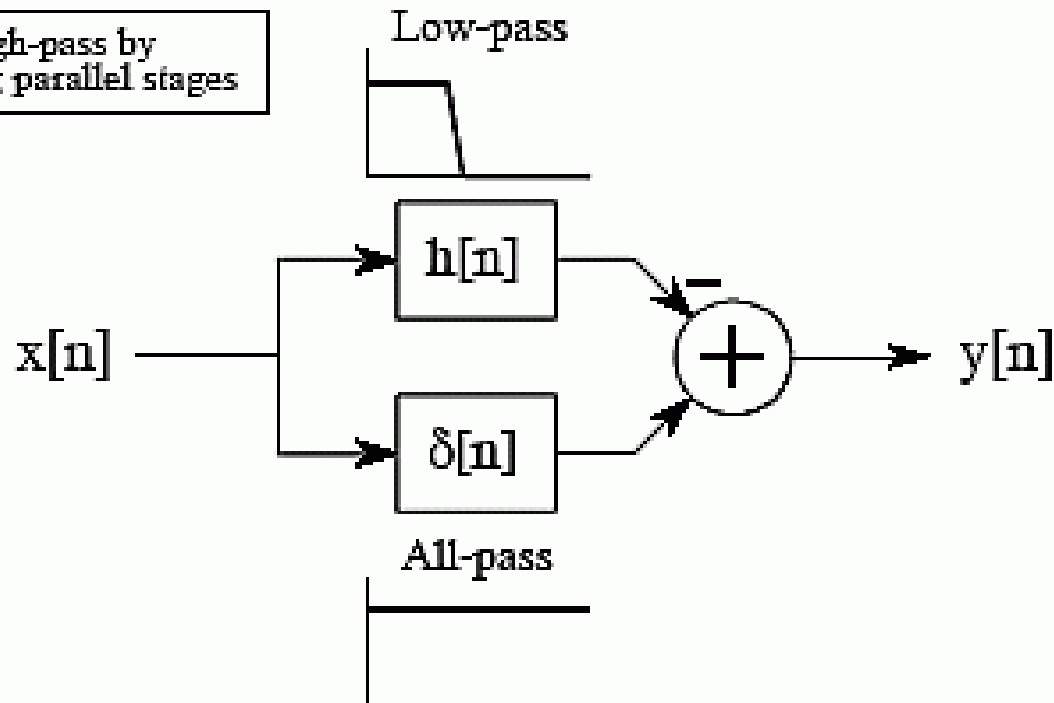


Ideal Filters

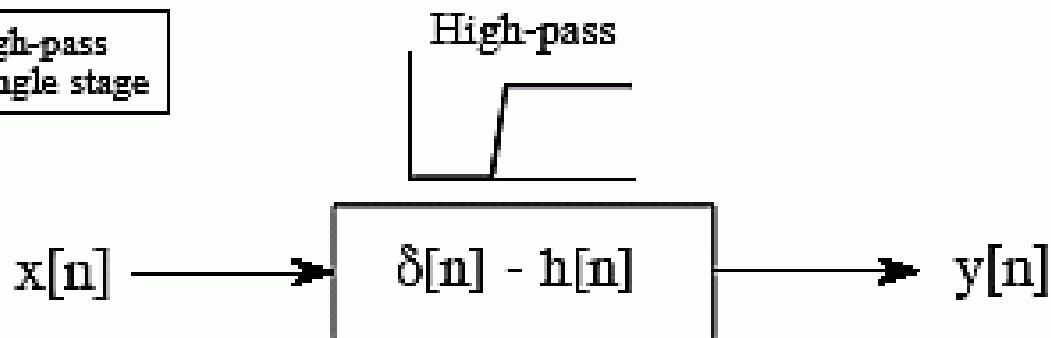


HP Digital Filter from LP Design

a. High-pass by adding parallel stages

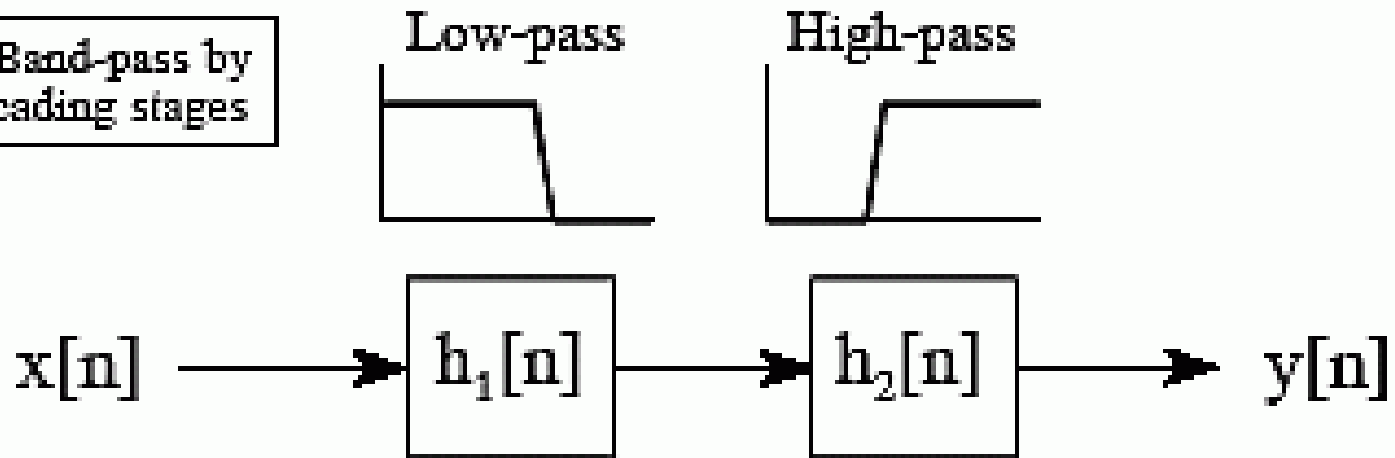


b. High-pass in a single stage

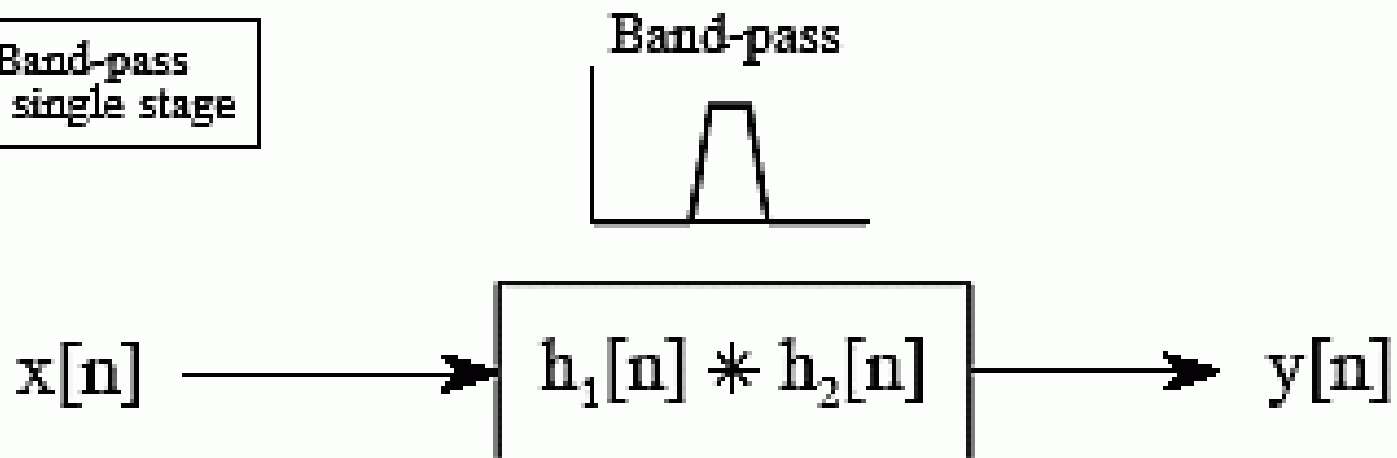


BP Digital Filter from HP & LP

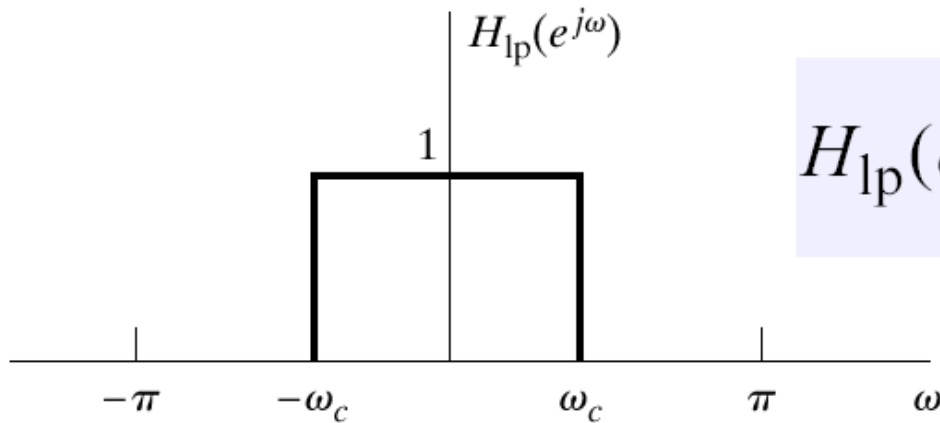
a. Band-pass by cascading stages



b. Band-pass in a single stage



Ideal Lowpass Filter



$$H_{lp}(e^{j\omega}) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

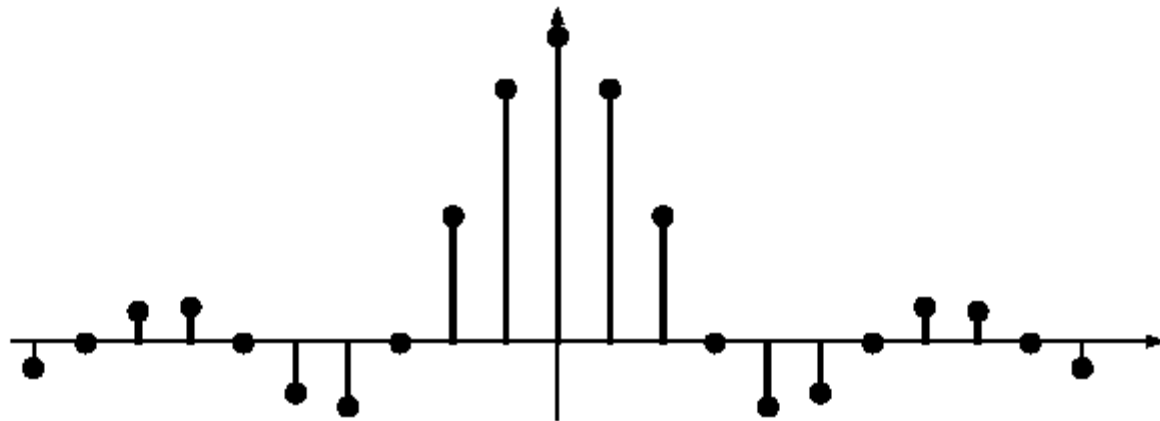
$$\begin{aligned} h_{lp}[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_{lp}(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega \\ &= \frac{1}{j2\pi n} \left[e^{j\omega n} \right]_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{j2\pi n} \end{aligned}$$

$$h_{lp}[n] = \frac{\sin \omega_c n}{\pi n} \quad -\infty < n < \infty$$

$h[n]$ of ideal filter

- $H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \text{elsewhere} \end{cases}, \quad -\pi \leq \omega \leq \pi$

- $h[n] = \frac{\sin \omega_c n}{\pi n}$

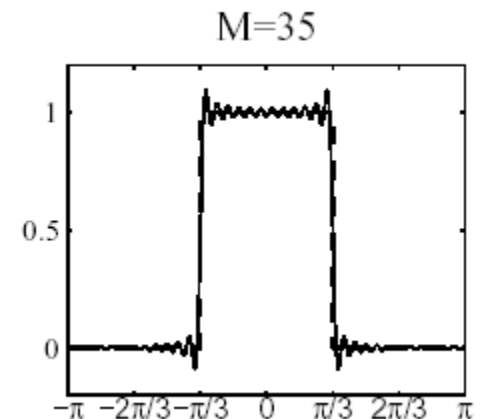
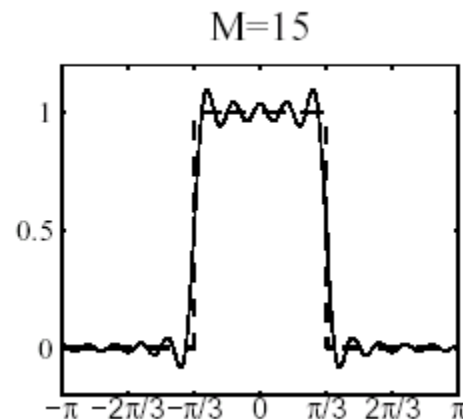
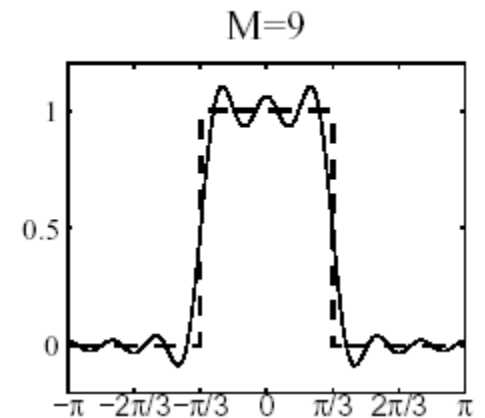
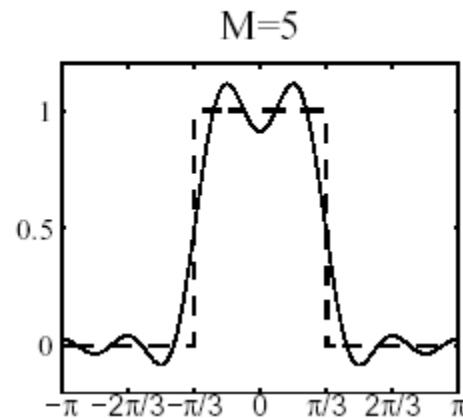


Approximations

- $h[n]$ IIR
- Approximation by an FIR filter:

$$H_M(e^{j\omega}) = \sum_{n=-M}^M h[n]e^{-j\omega n}$$

"Gibbs phenomenon"



Freq Axis

Relationship between continuous-time frequency Ω , discrete-time frequency ω , and DFT index k

$\Omega(\text{rad/s})$	$\omega(\text{rad})$	k
0	0	0
$\Omega_s/2$	π	$N/2$
Ω_s	2π	N^*

The highest discrete-time frequency is $\omega = \pi$,
 $\omega = 2\pi$ is the same as $\omega = 0$, the lowest frequency.

Inverse System

- An inverse system compensates (undoes) the effects of another system.

